Tempest: Porewater Nutrients

2025 Samples

2026-05-27

Contents

0.1 Run Information

```
cat("Run Information: NAME ") #lets you know what section you're in
```

```
## Run Information: NAME
```

```
#set the run date & user name
  sample_year <- "2024|2025|2026"
  user <- "Brenden Blakley"

#identify the files you want to read in
  #read in as a list to accommadate ultiple runs in a month
NH3_PO4_files<- c("Raw Data/20260504_TMP2025_NP_Run3.csv", "Raw Data/20260508_TMPmix_NP_Run1.csv")
  NOx_files    <- c("Raw Data/20260504_TMP2025_NOx_Run3.csv", "Raw Data/20260508_TMPmix_NOx_Run1.csv")

# Define the file path for QAQC log file - NO Need to change just check year
  file_path <- "Raw Data/SEAL_TEMPEST_QAQC_Log_2024.csv"
  final_path <- "Processed Data/Tempest_Nutrients_2025_Run3_Runmix.csv"

#record any notes about the run or anything other info here:
  run_notes <- ""

#Set up file path for metadata
  #downloaded metadata csv - downloaded from Google drive as csv for this year
  Raw_Metadata = "Raw Data/COMPASS_SynopticCB_PW_SampleLog_2024.csv"

  cat(run_notes)
```

##Setup

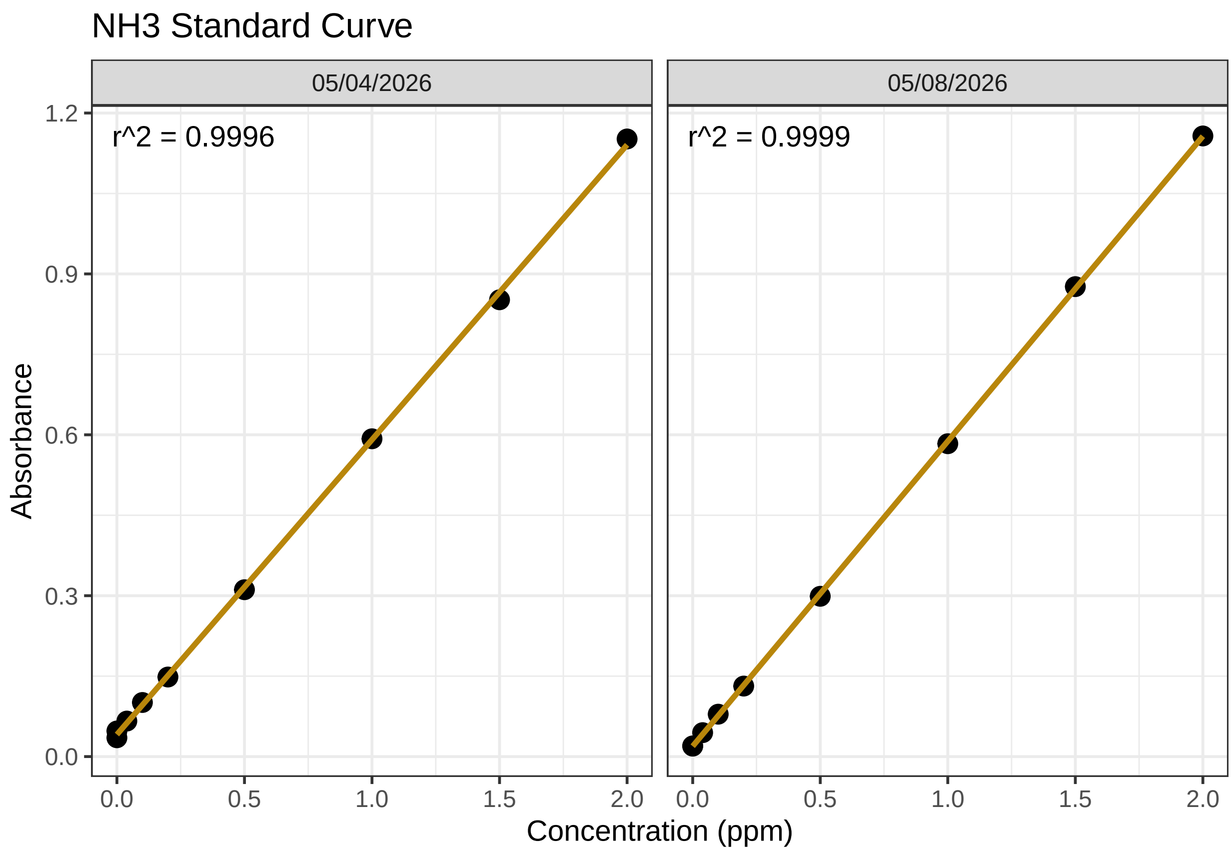
##Pull in active porewater tracking inventory sheet from Google Drive:

##Create similar sample IDs to match with run samples

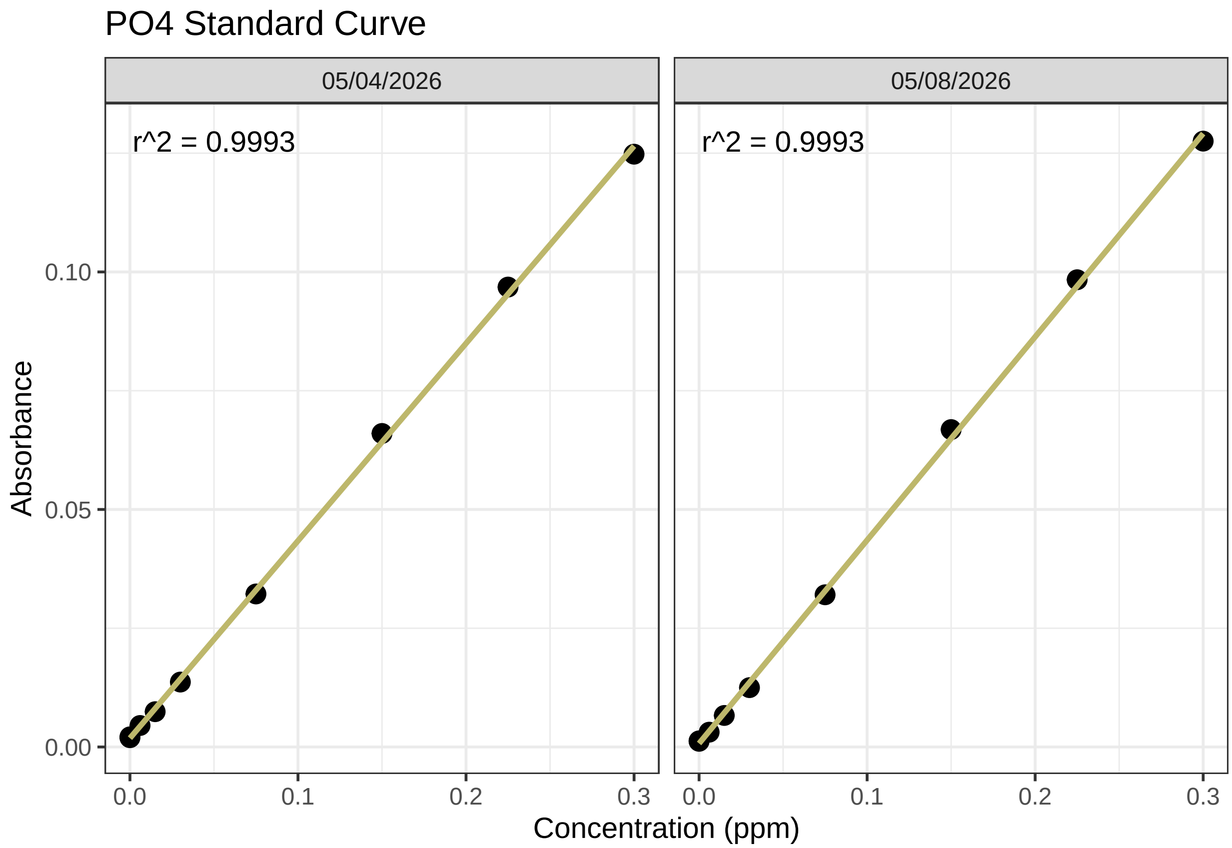
##Import Data & Clean

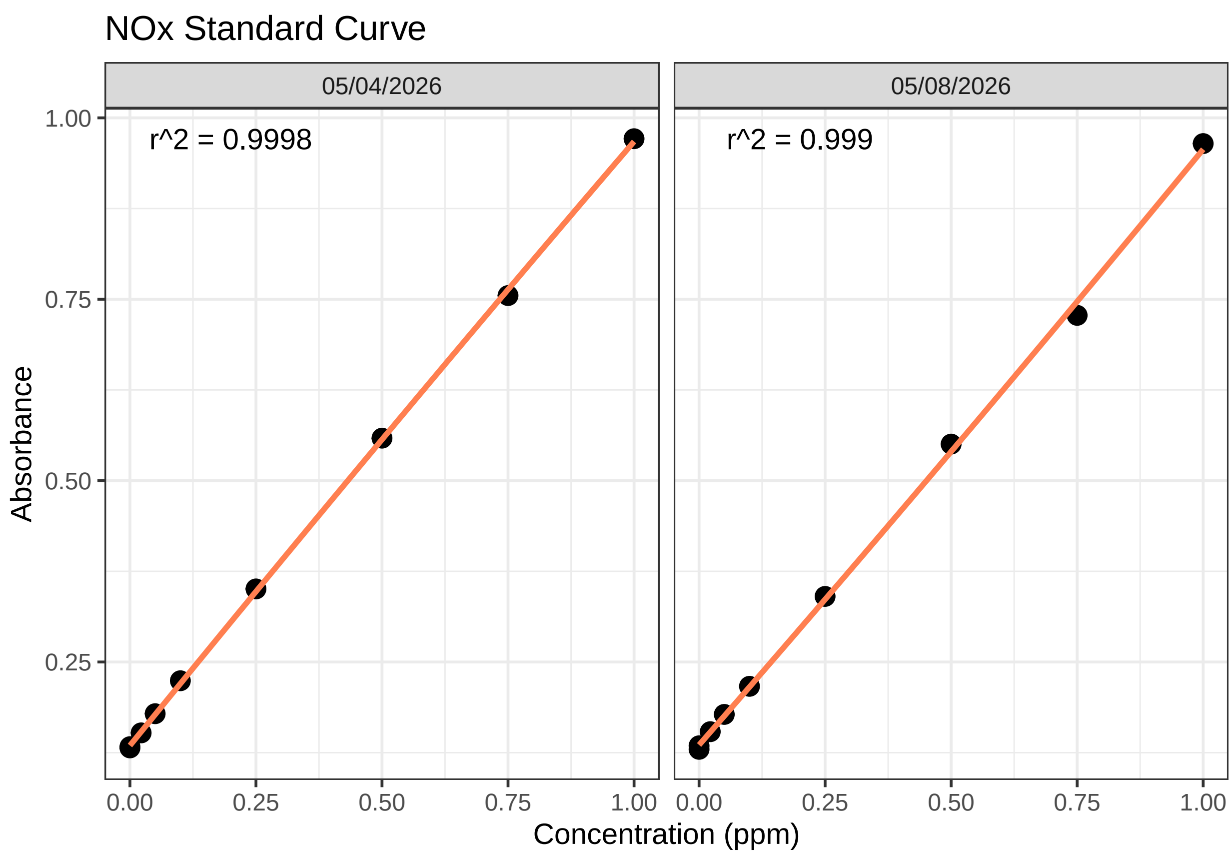
0.2 Assessing standard Curves

```
## `geom_smooth()` using formula = 'y ~ x'
```



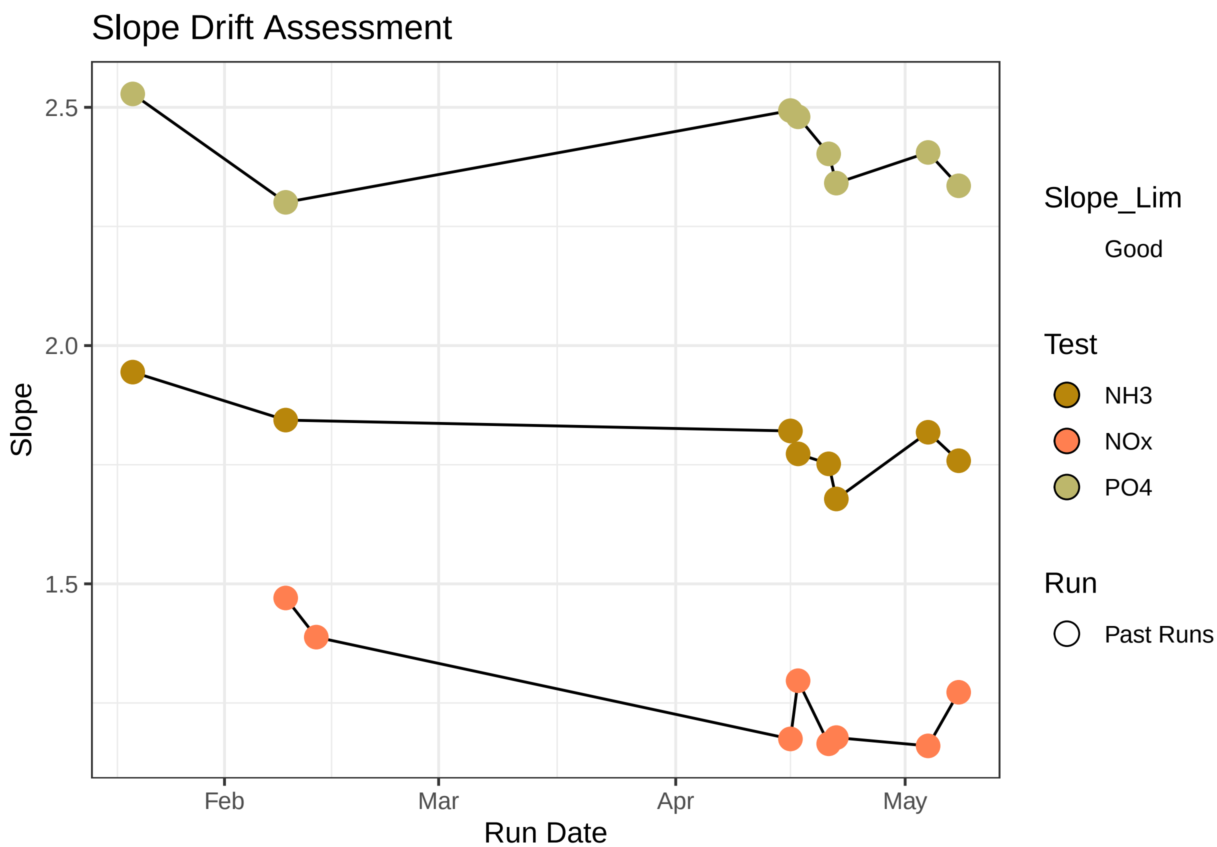

```
## `geom_smooth()` using formula = 'y ~ x'
```

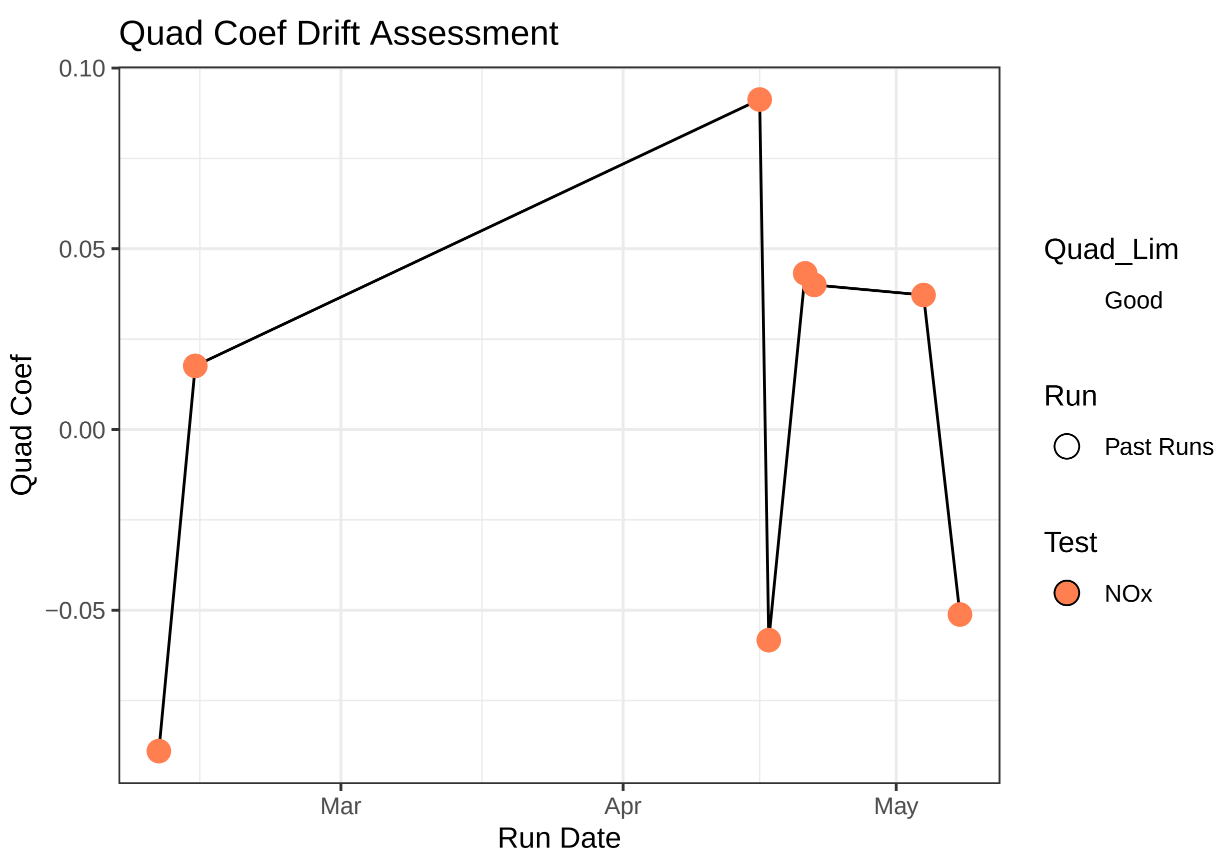

PO4 Standard Curve
05/04/2026
05/08/2026
r^2 = 0.9993
r^2 = 0.9993
Absorbance
0.10
0.05
0.00
0.0
0.1
0.2
0.3
Concentration (ppm)

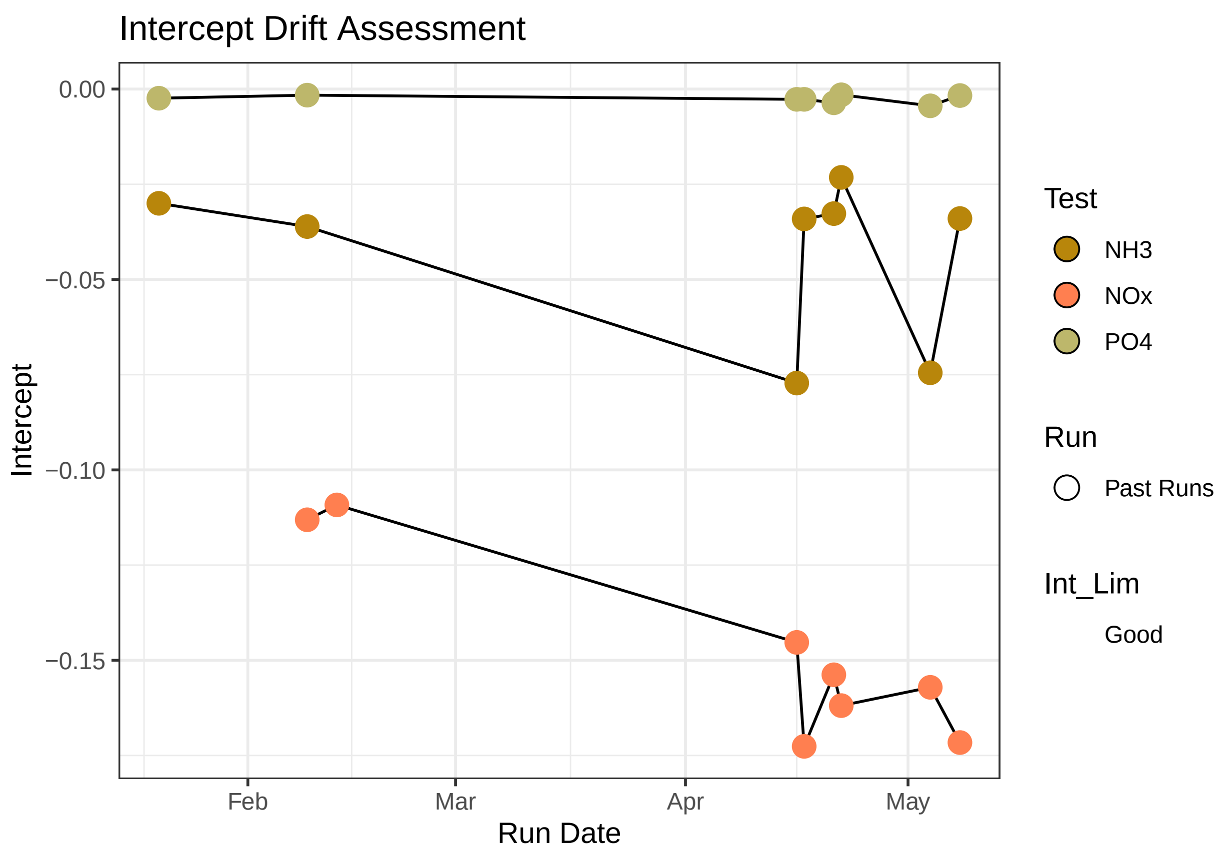


Test	Run_Date	R2_Flag	R2	Slope
Ammonia 2	05/04/2026	R2_Pass	0.9995982	1.818107
Ammonia 2	05/08/2026	R2_Pass	0.9999344	1.758303
o-PHOS 0.3	05/04/2026	R2_Pass	0.9993257	2.405344
o-PHOS 0.3	05/08/2026	R2_Pass	0.9993370	2.335316
Vanadium NOx	05/04/2026	R2_Pass	0.9997660	1.159949
Vanadium NOx	05/08/2026	R2_Pass	0.9989603	1.272420

0.3 Update Standard Log






Intercept Drift Assessment
Intercept
0.00
−0.05
−0.10
−0.15
Feb
Mar
Apr
May
Run Date
Test
NH3
NOx
PO4
Run
Past Runs
Int_Lim
Good

0.4 Dilution Corrections - ensure the latest dilution is kept

```
## [1] "Reran Samples: TMP_SW_D5_20251101"
```

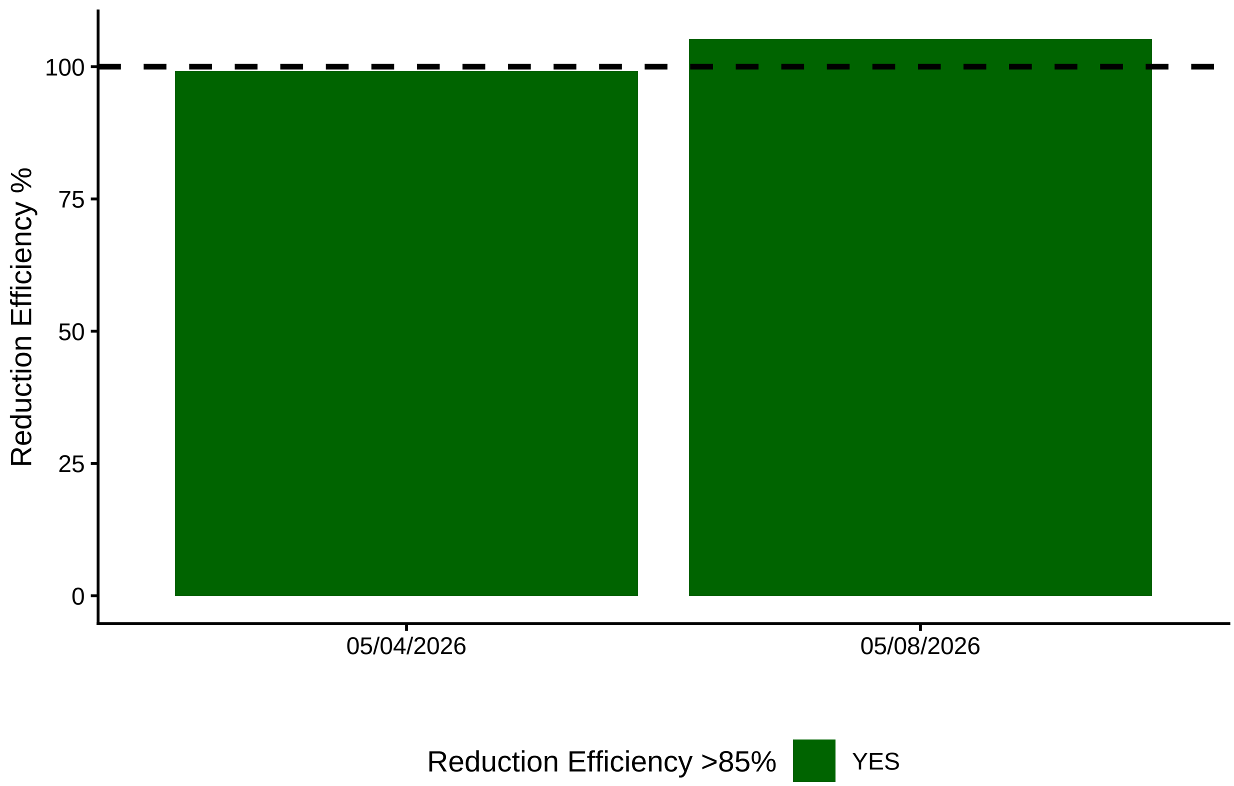
```
## [1] "Diluted Samples: TMP_SW_D5_20251101"
```

```
## [1] "No Naming Issues Detected"
```

0.5 Performance Check

Test	Run_Date	PE_Flag	PE_Conc	PE_Target_Conc
Ammonia 2	05/04/2026	Performance Check Within 25% - PROCEED	1.2530565	1.034
Ammonia 2	05/08/2026	Performance Check Within 25% - PROCEED	1.2651620	1.034
o-PHOS 0.3	05/04/2026	Performance Check Within 25% - PROCEED	0.9222755	0.824
o-PHOS 0.3	05/08/2026	Performance Check Within 25% - PROCEED	0.9105505	0.824
Vanadium NOx	05/04/2026	Performance Check Within 25% - PROCEED	1.5966770	1.510
Vanadium NOx	05/08/2026	Performance Check Within 25% - PROCEED	1.6844615	1.510

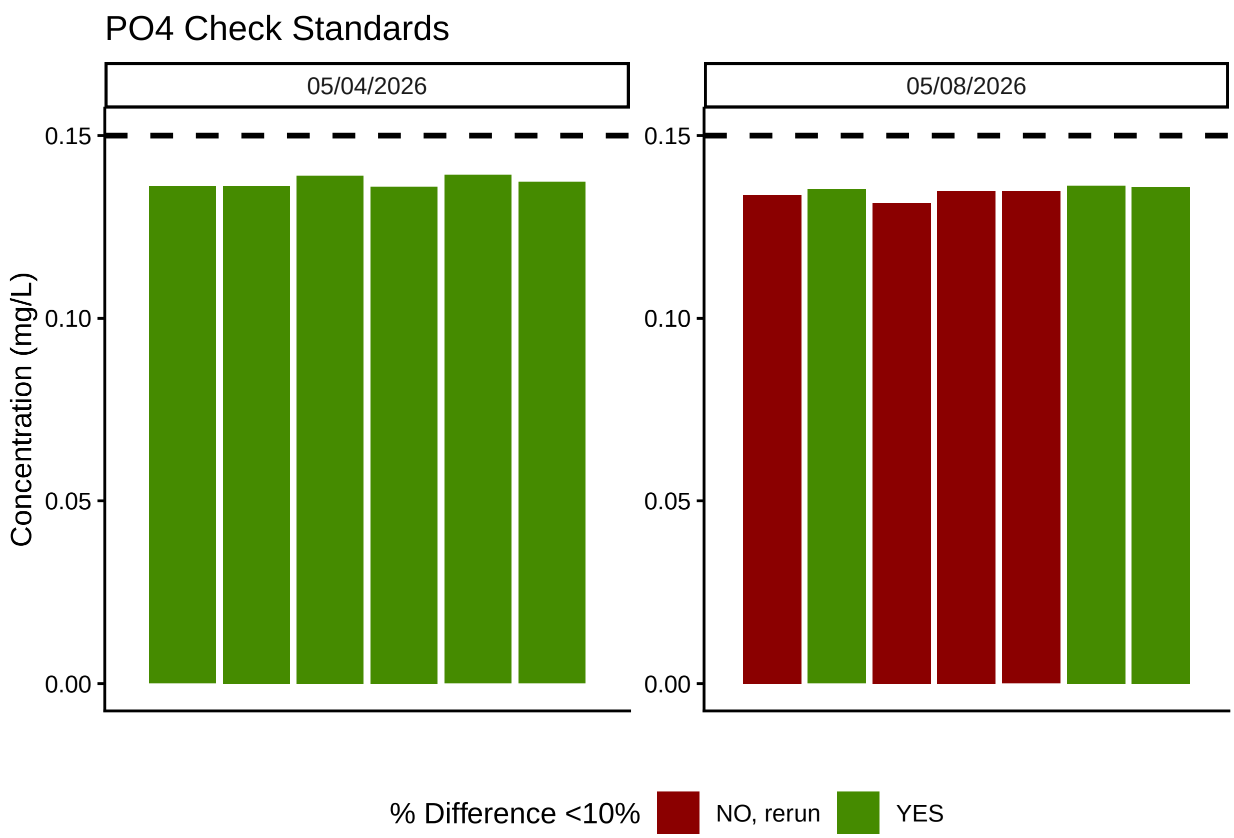
0.6 Check NOx Reduction Efficiency

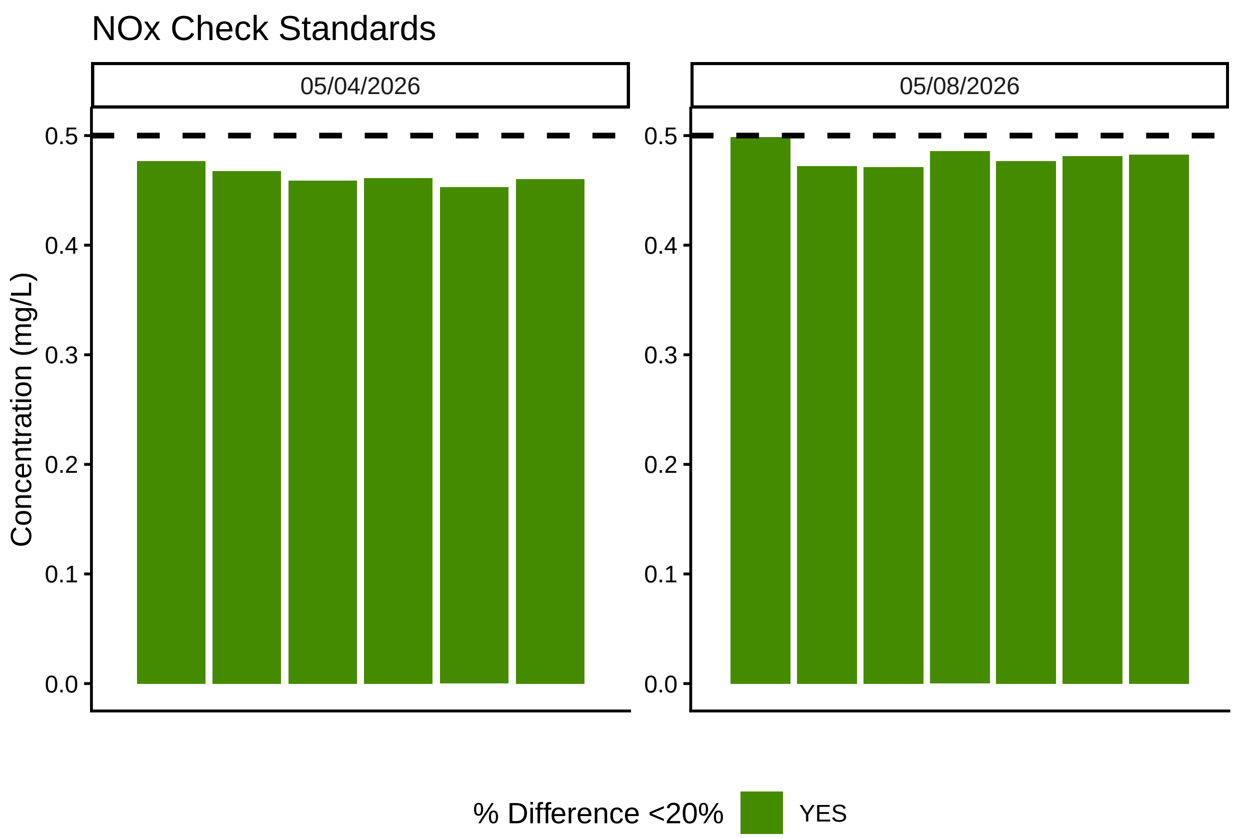


Test	Run_Date	Red_Eff_Flag
	05/04/2026	Mean NOx Reduction Efficiency >85% - PROCEED
	05/08/2026	Mean NOx Reduction Efficiency >85% - PROCEED

0.7 Analyze the Check Standards

Test	Run_Date	RSV_Flag	RSV	RSV_Cutoff
Ammonia 2	05/04/2026	RSV WITHIN RANGE - PROCEED	0.0207134	0.25
Ammonia 2	05/08/2026	RSV WITHIN RANGE - PROCEED	0.0207134	0.25
o-PHOS 0.3	05/04/2026	RSV WITHIN RANGE - PROCEED	0.0150140	0.25
o-PHOS 0.3	05/08/2026	RSV WITHIN RANGE - PROCEED	0.0150140	0.25
Vanadium NOx	05/04/2026	RSV WITHIN RANGE - PROCEED	0.0269662	0.25
Vanadium NOx	05/08/2026	RSV WITHIN RANGE - PROCEED	0.0269662	0.25

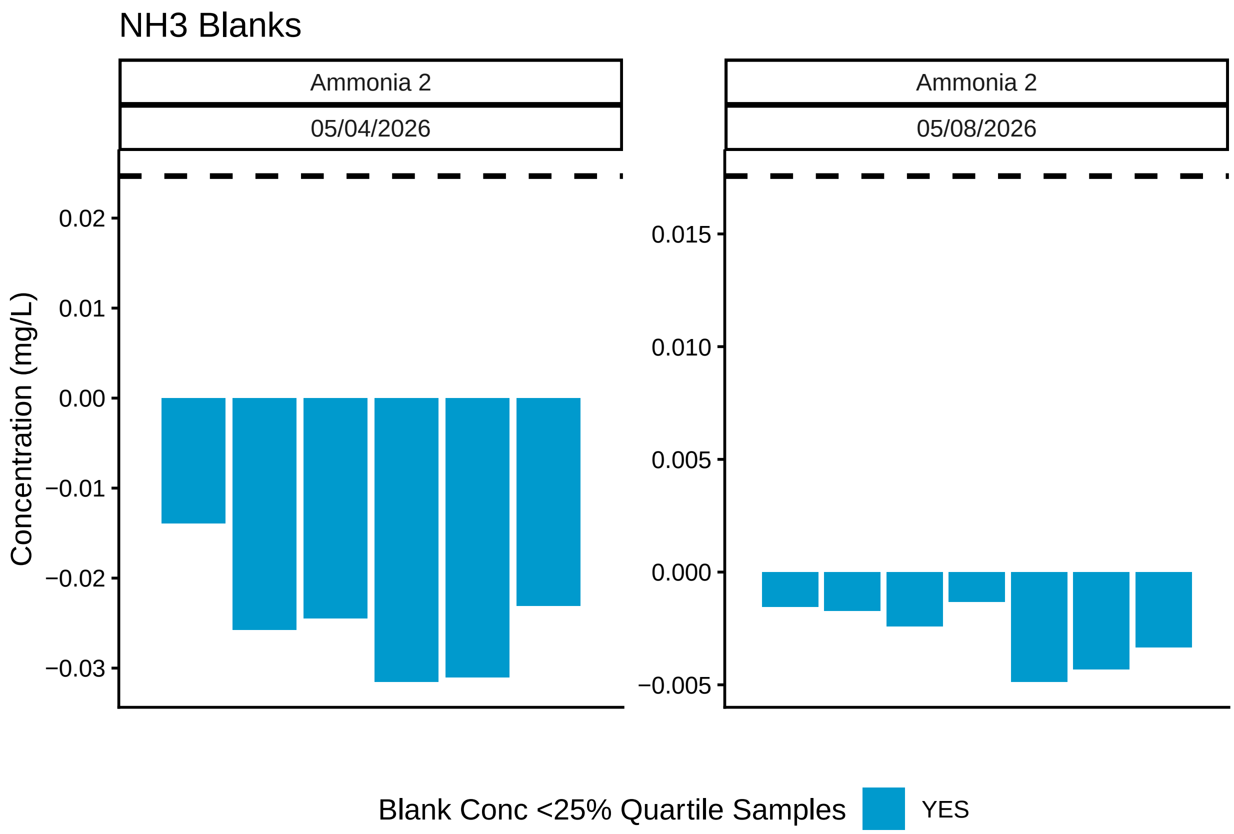

PO4 Check Standards
05/04/2026
05/08/2026
0.15
0.10
0.05
0.00
Concentration (mg/L)
% Difference <10%
NO, rerun
YES

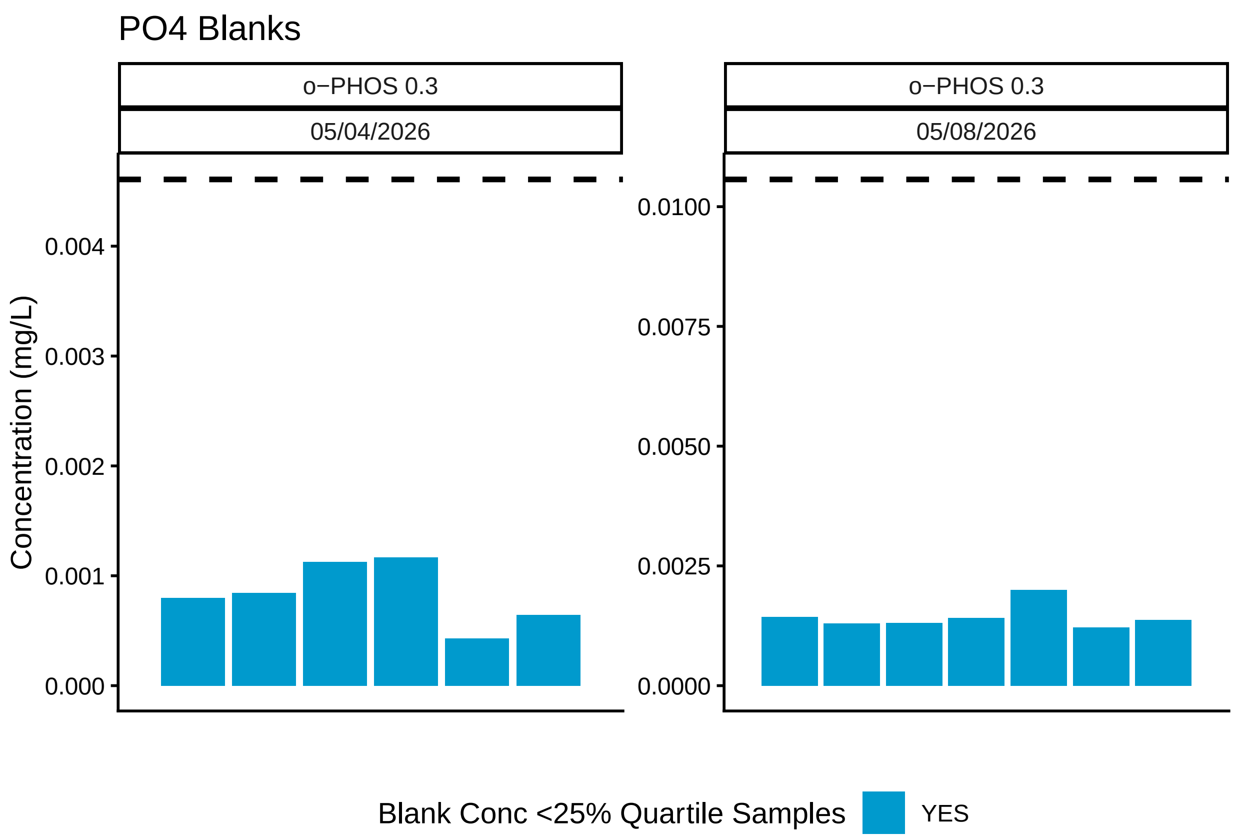


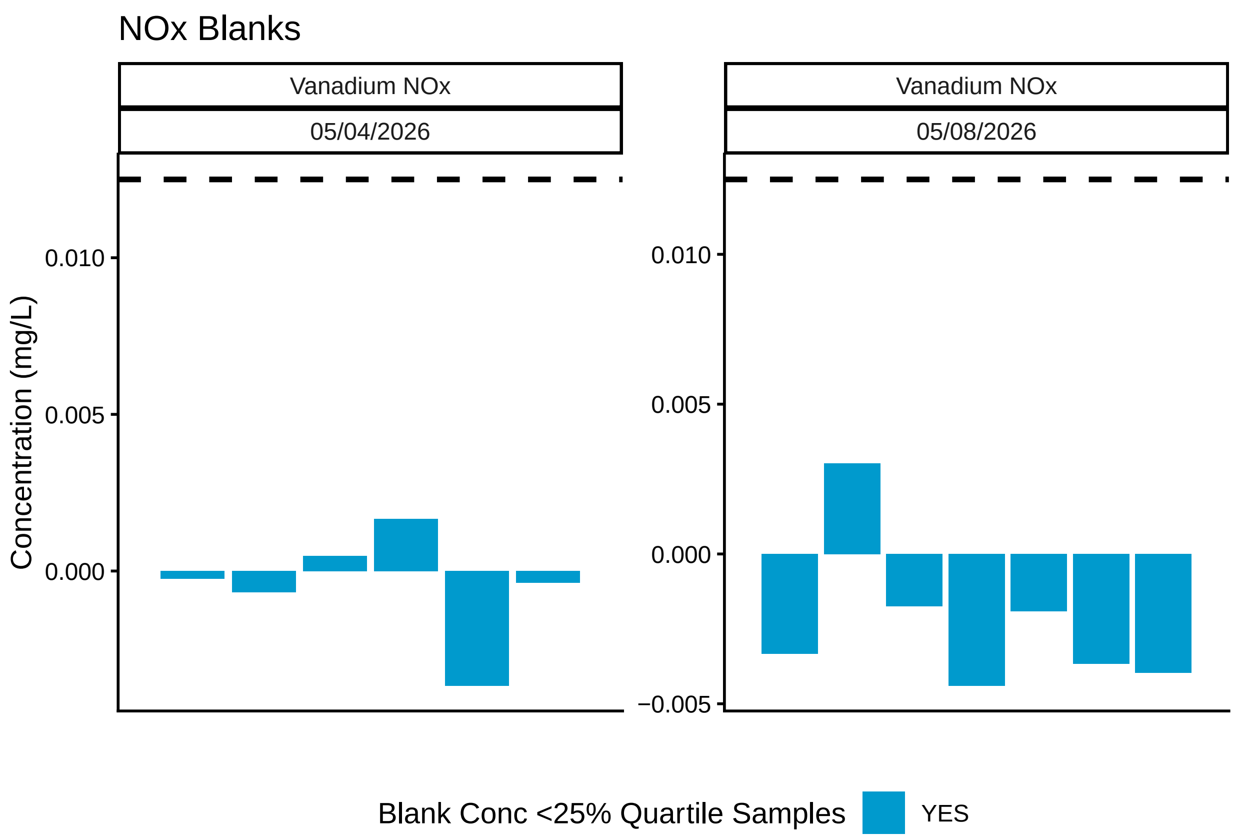
Test	Run_Date	CHK_Flag
Ammonia 2	05/04/2026	>60% of Checks Pass - PROCEED
Ammonia 2	05/08/2026	>60% of Checks Pass - PROCEED
o-PHOS 0.3	05/04/2026	>60% of Checks Pass - PROCEED
o-PHOS 0.3	05/08/2026	<60% of Checks Pass - REASSESs
Vanadium NOx	05/04/2026	>60% of Checks Pass - PROCEED
Vanadium NOx	05/08/2026	>60% of Checks Pass - PROCEED

0.8 Analyze Blanks

Test	Run_Date	BLK_Pct_Flag	Mean_Blk_Conc	Quantile_25
Ammonia 2	05/04/2026	>60% of Blanks Pass - PROCEED	-0.0249550	0.0246705
Ammonia 2	05/08/2026	>60% of Blanks Pass - PROCEED	-0.0027873	0.0175657
o-PHOS 0.3	05/04/2026	>60% of Blanks Pass - PROCEED	0.0008352	0.0046075
o-PHOS 0.3	05/08/2026	>60% of Blanks Pass - PROCEED	0.0014364	0.0105687
Vanadium NOx	05/04/2026	>60% of Blanks Pass - PROCEED	-0.0004657	0.0125000
Vanadium NOx	05/08/2026	>60% of Blanks Pass - PROCEED	-0.0022839	0.0125000

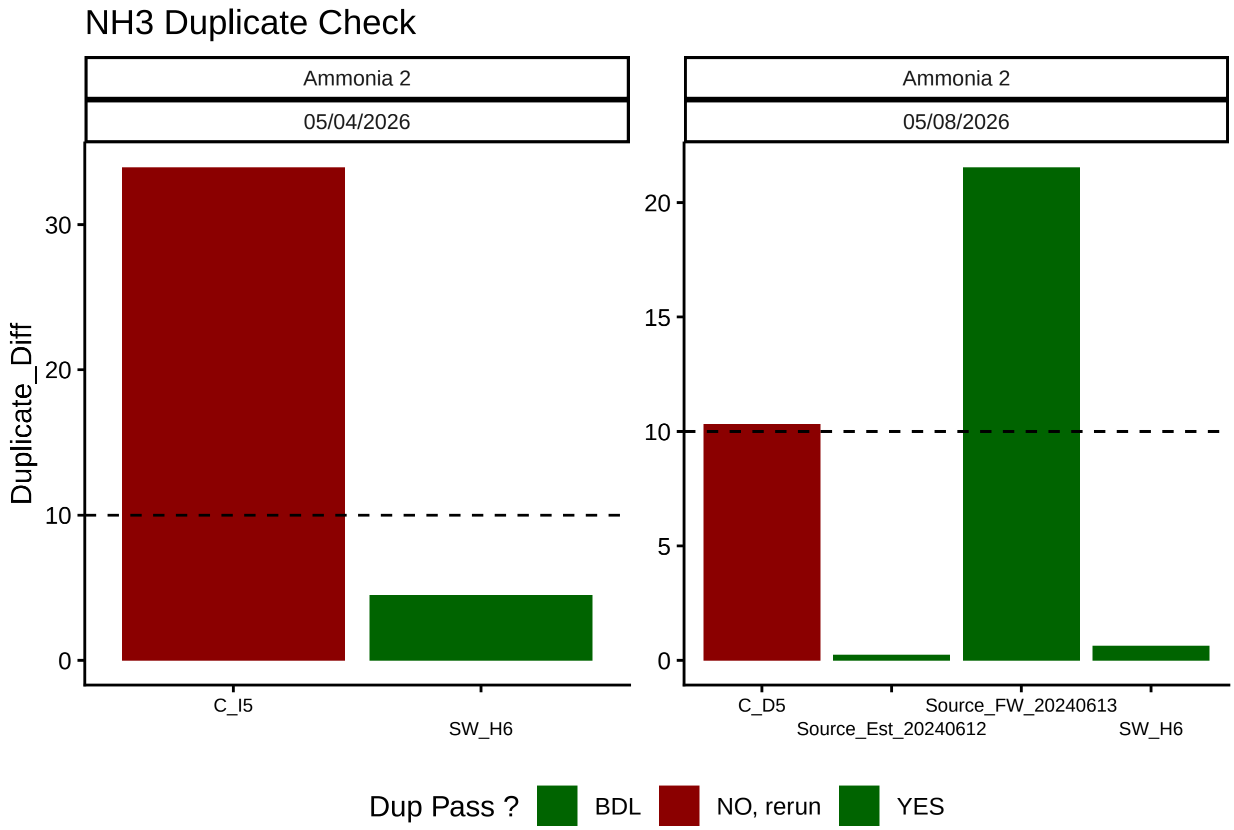


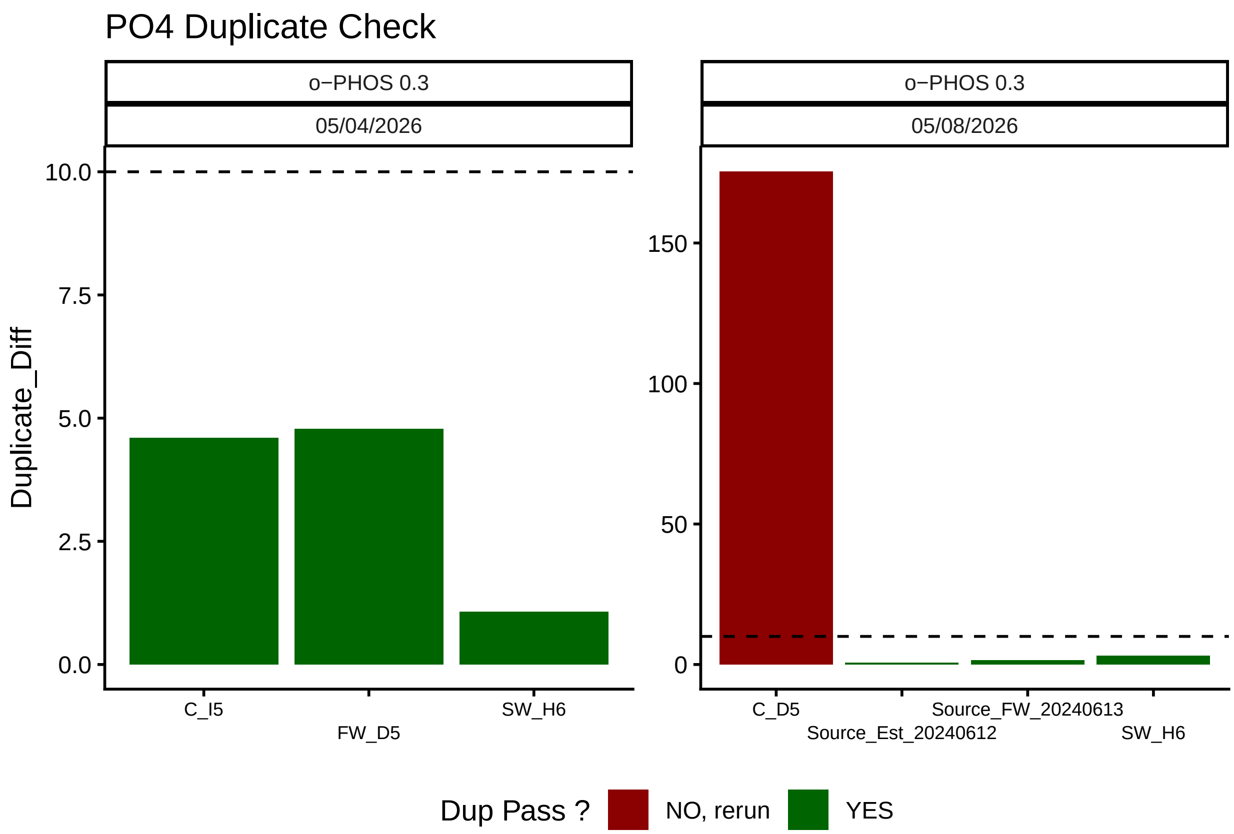

PO4 Blanks
o−PHOS 0.3
05/04/2026
o−PHOS 0.3
05/08/2026
Concentration (mg/L)
0.004
0.003
0.002
0.001
0.000
0.0100
0.0075
0.0050
0.0025
0.0000
Blank Conc <25% Quartile Samples
YES

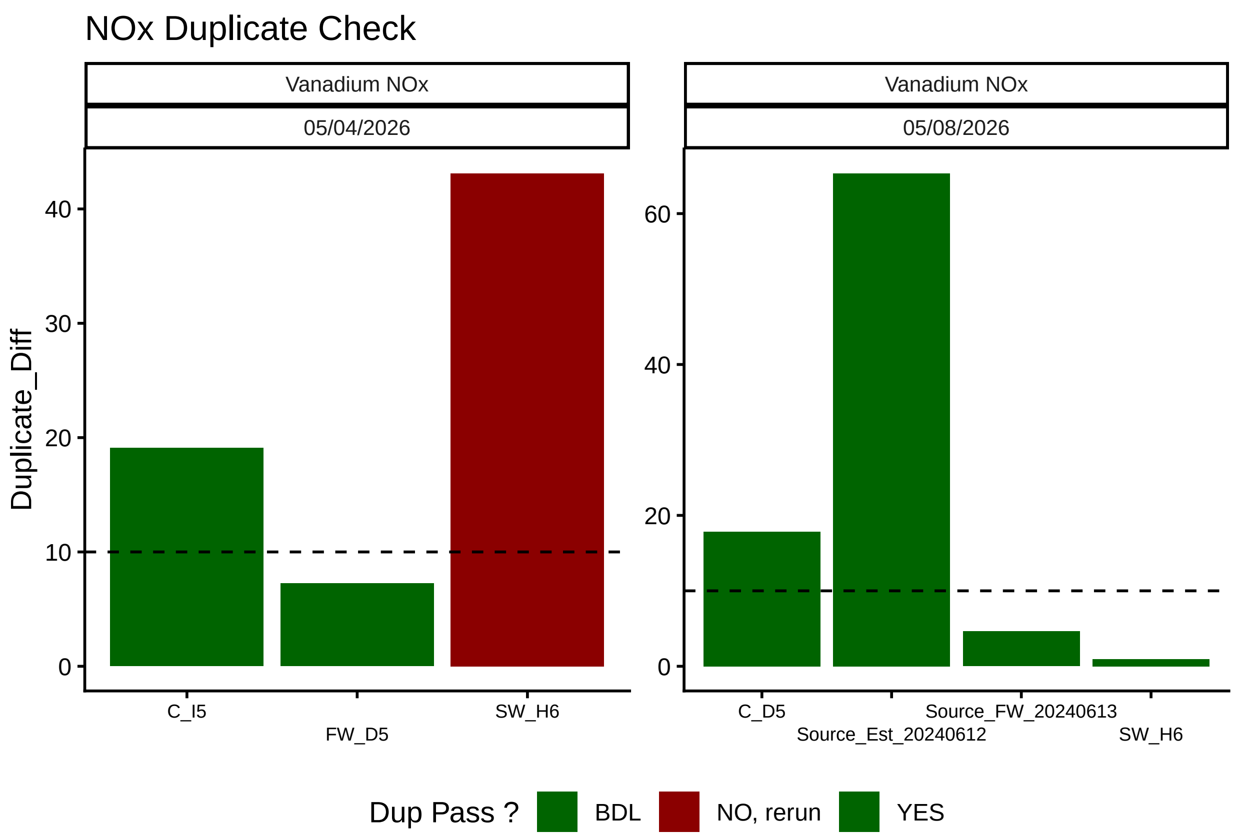


NOx Blanks
Vanadium NOx
05/04/2026
Vanadium NOx
05/08/2026
Concentration (mg/L)
0.010
0.005
0.000
0.010
0.005
0.000
−0.005
Blank Conc <25% Quartile Samples
YES

0.9 Analyze Duplicates

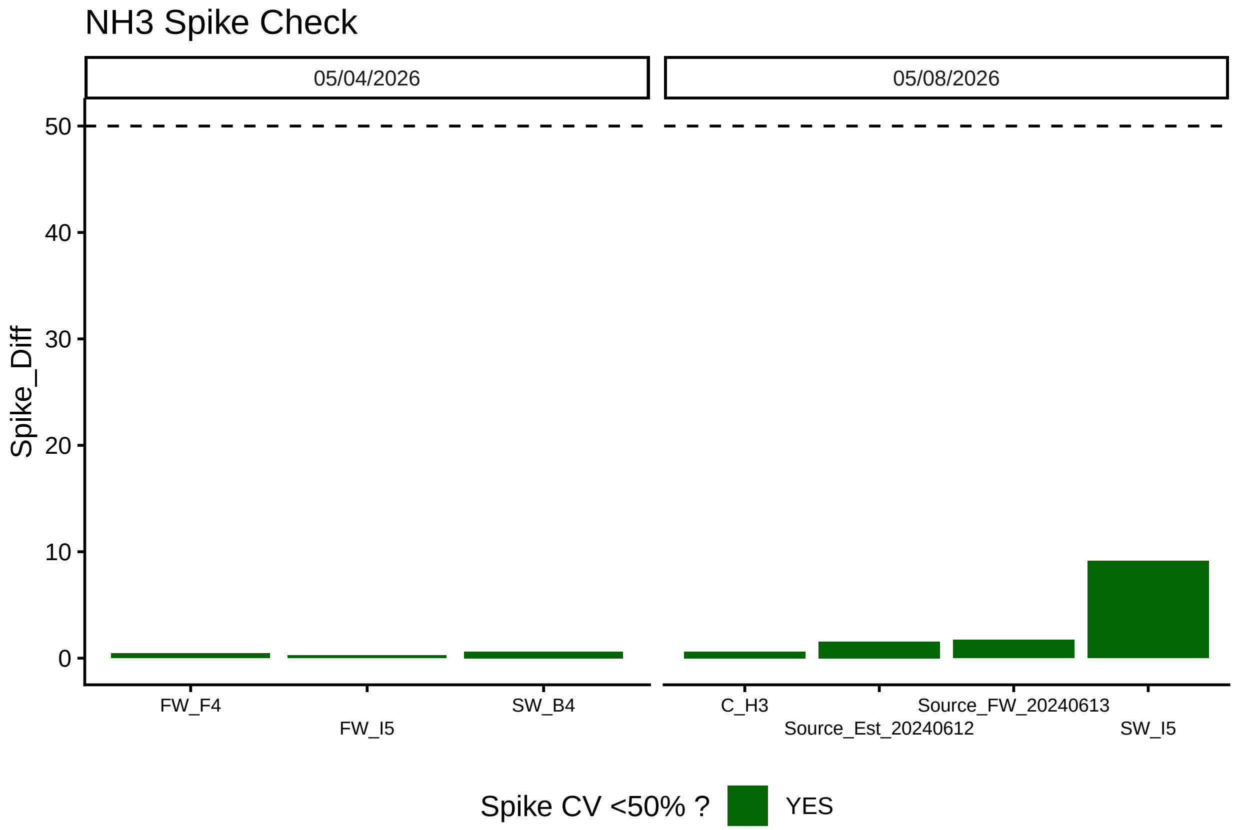


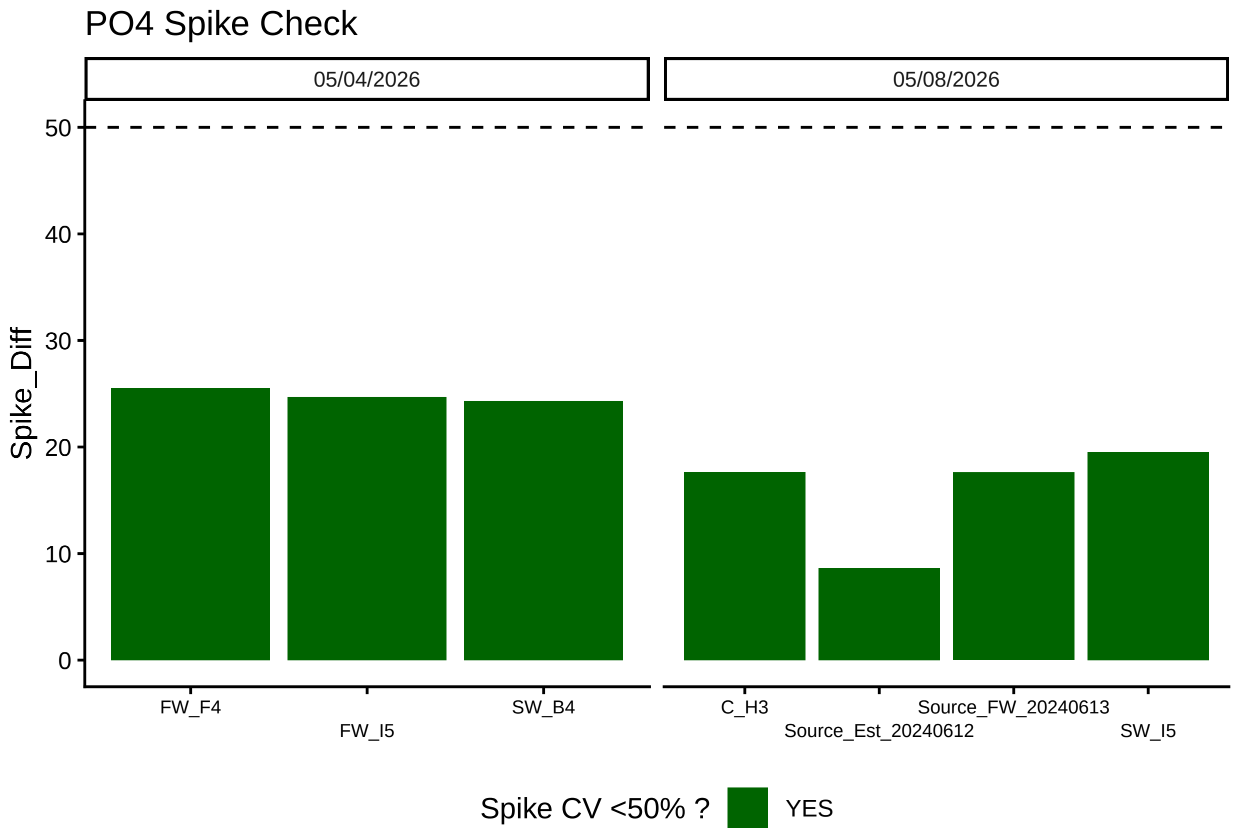


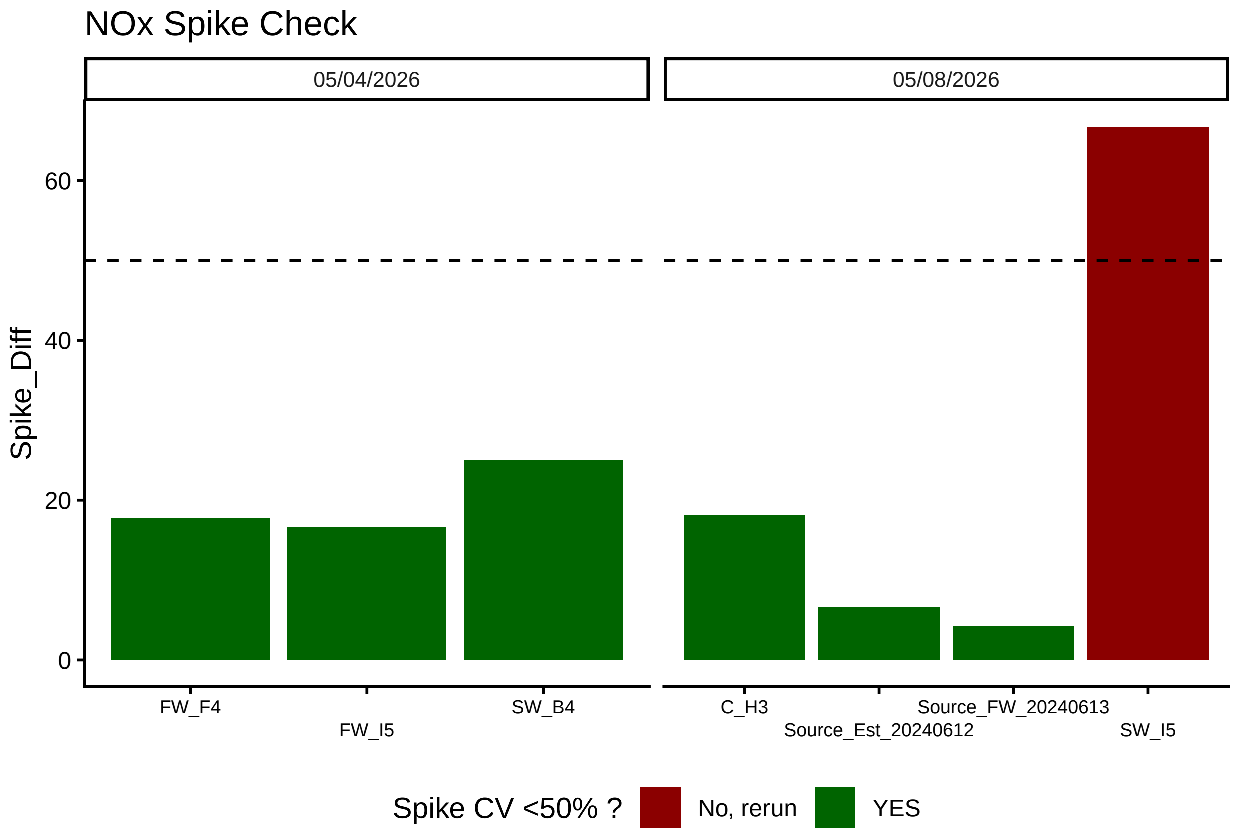


Test	Run_Date	Dup_Flags
Ammonia 2	05/04/2026	<60% of Dups Pass - REASSESS
Ammonia 2	05/08/2026	<60% of Dups Pass - REASSESS
o-PHOS 0.3	05/04/2026	>60% of Dups Pass - PROCEED
o-PHOS 0.3	05/08/2026	>60% of Dups Pass - PROCEED
Vanadium NOx	05/04/2026	<60% of Dups Pass - REASSESS
Vanadium NOx	05/08/2026	<60% of Dups Pass - REASSESS

0.10 Spikes

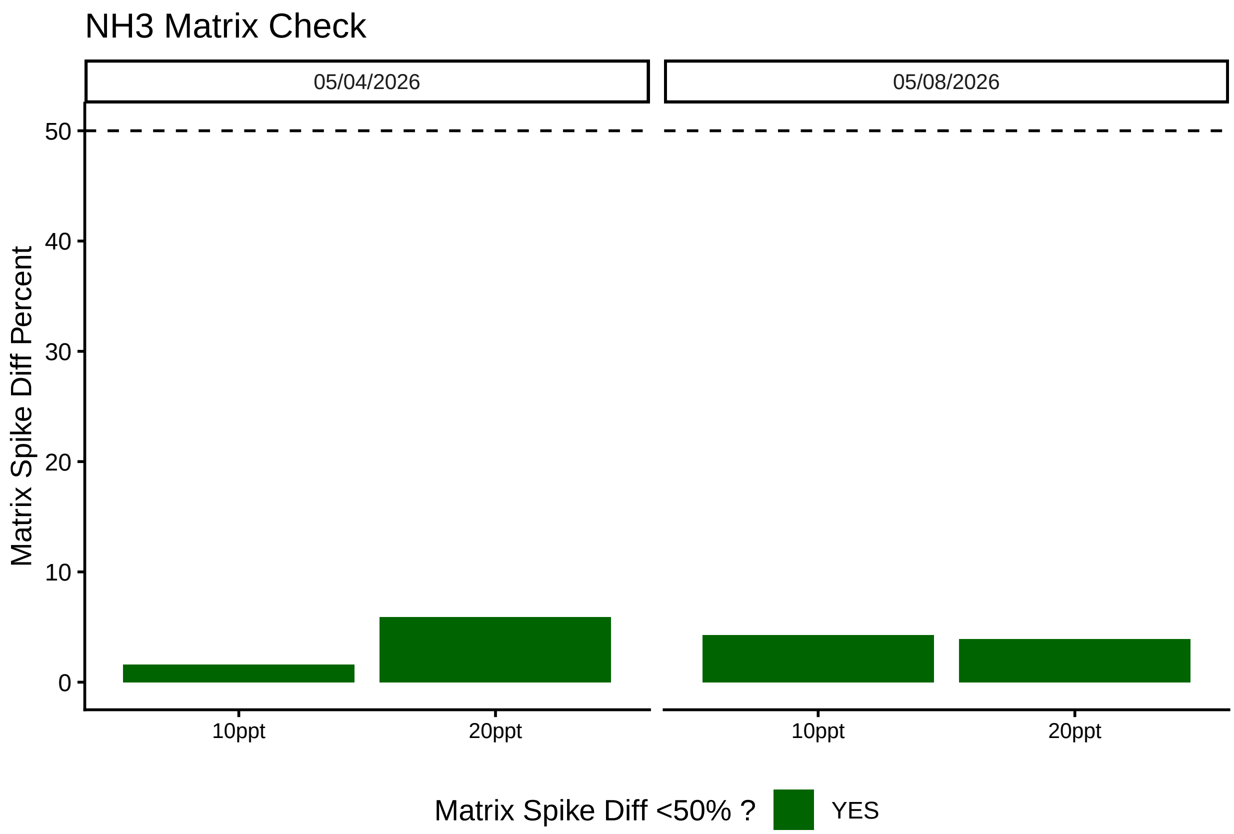

NH3 Spike Check
05/04/2026
05/08/2026
Spike_Diff
0
10
20
30
40
50
FW_F4
FW_I5
SW_B4
C_H3
Source_Est_20240612
Source_FW_20240613
SW_I5
Spike CV <50% ?
YES

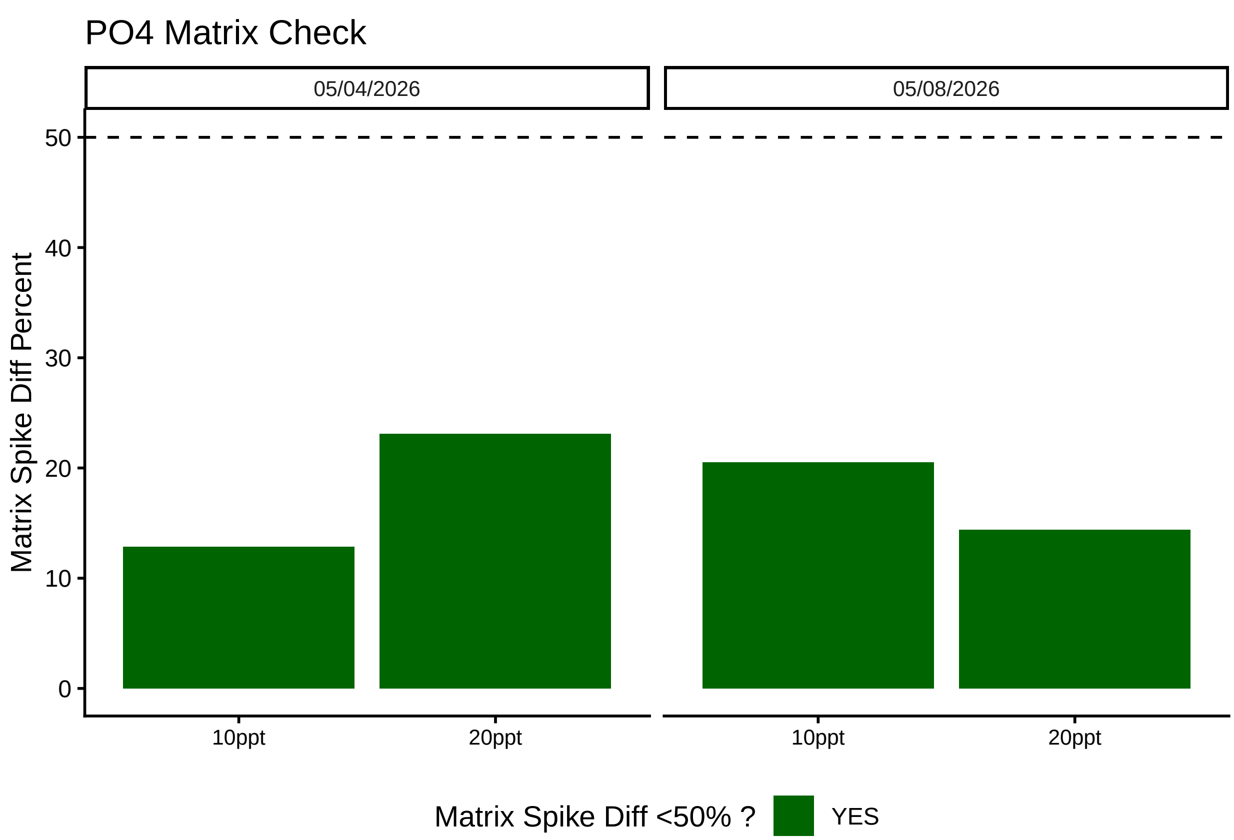

PO4 Spike Check
05/04/2026
05/08/2026
Spike_Diff
0
10
20
30
40
50
FW_F4
FW_I5
SW_B4
C_H3
Source_Est_20240612
Source_FW_20240613
SW_I5
Spike CV <50% ?
YES

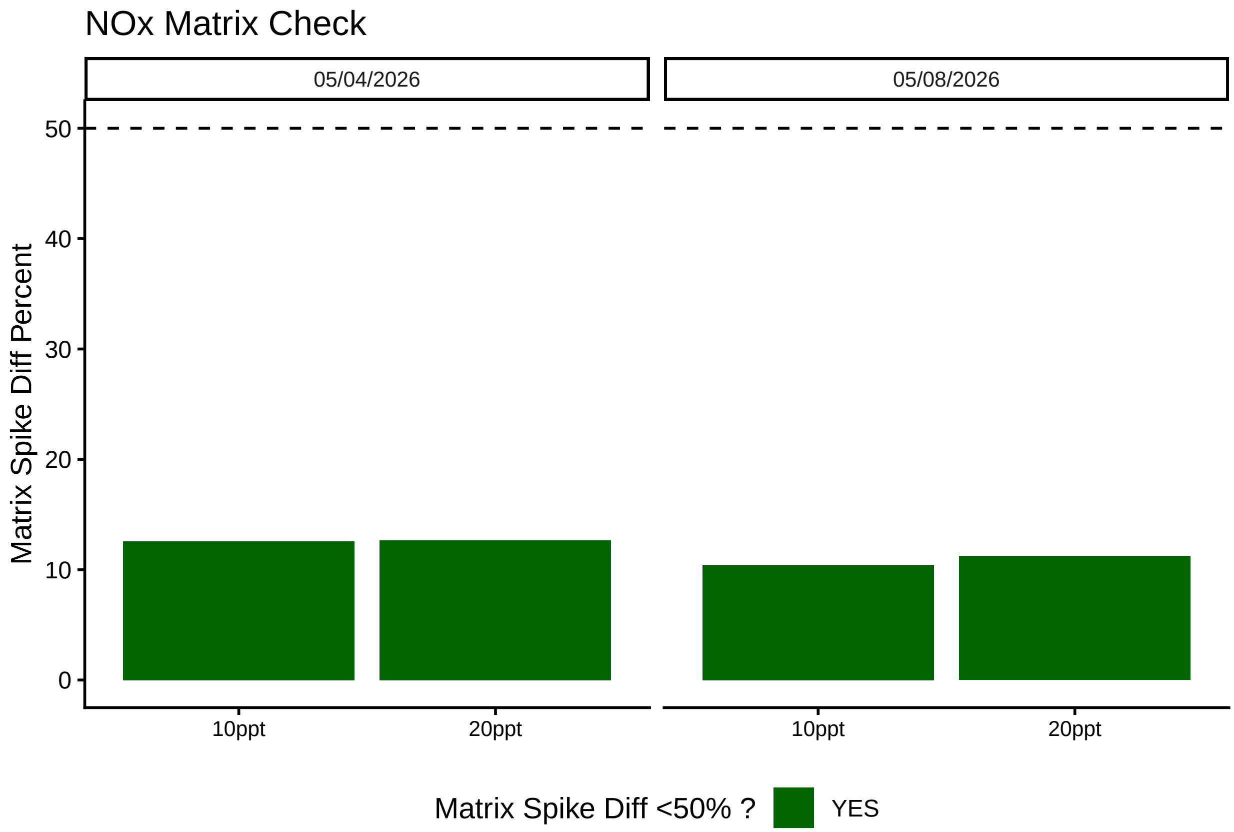


Test	Run_Date	Spike_Flags
Ammonia 2	05/04/2026	>60% of Spikes have a Diff <50% - PROCEED
Ammonia 2	05/08/2026	>60% of Spikes have a Diff <50% - PROCEED
Vanadium NOx	05/04/2026	>60% of Spikes have a Diff <50% - PROCEED
Vanadium NOx	05/08/2026	>60% of Spikes have a Diff <50% - PROCEED
o-PHOS 0.3	05/04/2026	>60% of Spikes have a Diff <50% - PROCEED
o-PHOS 0.3	05/08/2026	>60% of Spikes have a Diff <50% - PROCEED

0.11 Matrix Effects

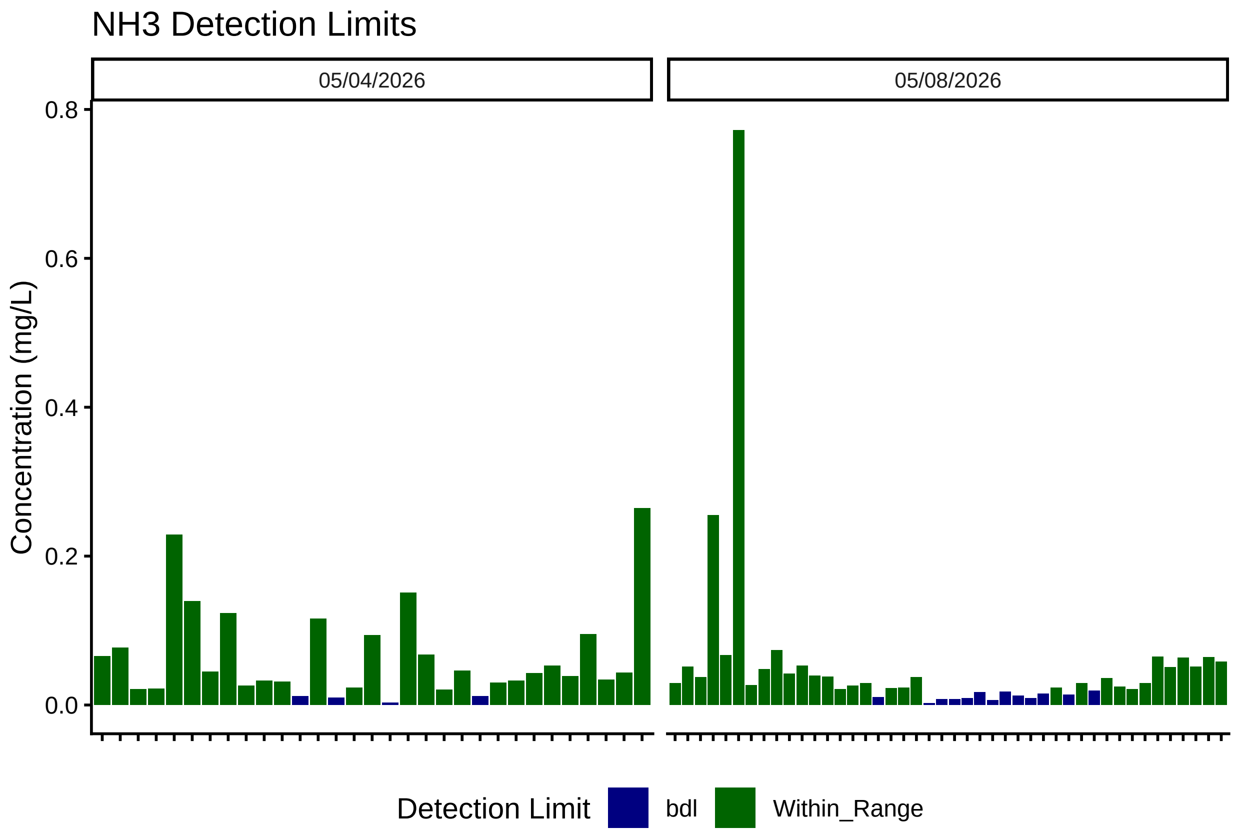


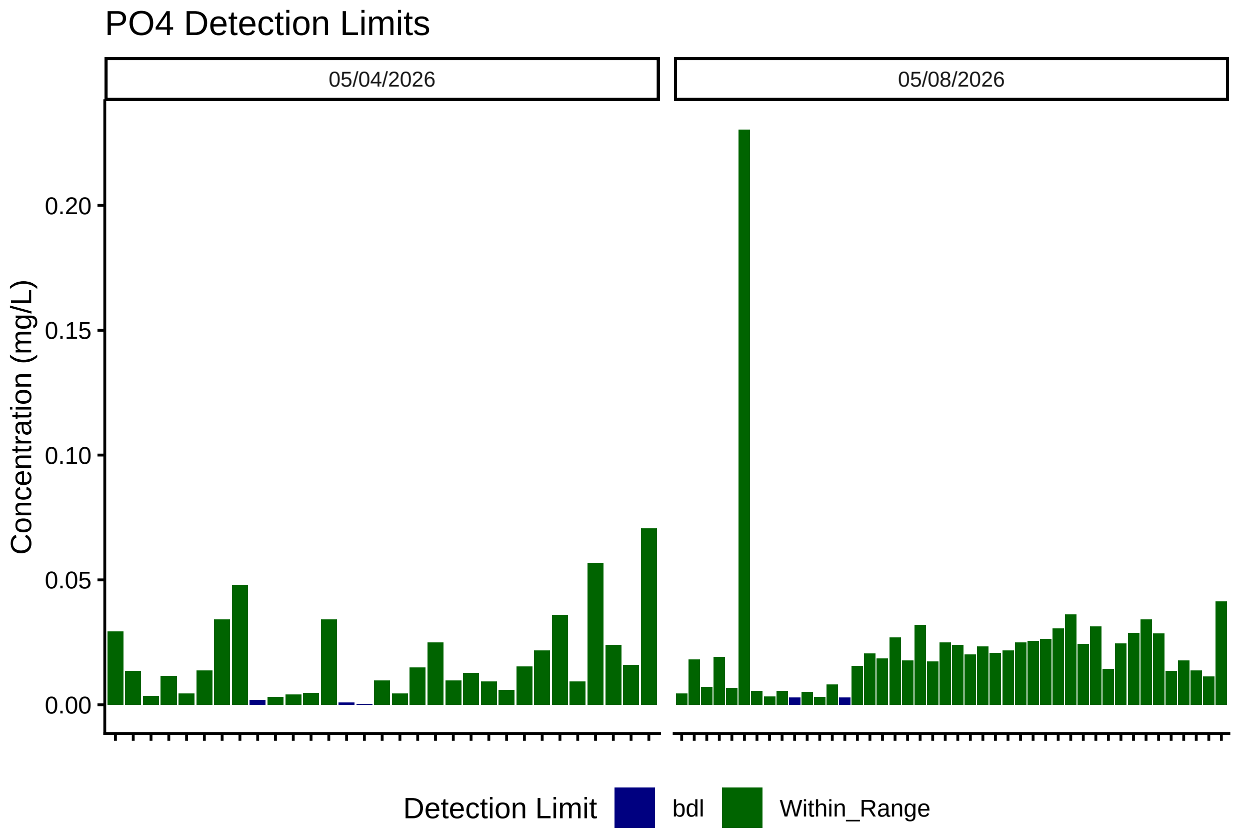


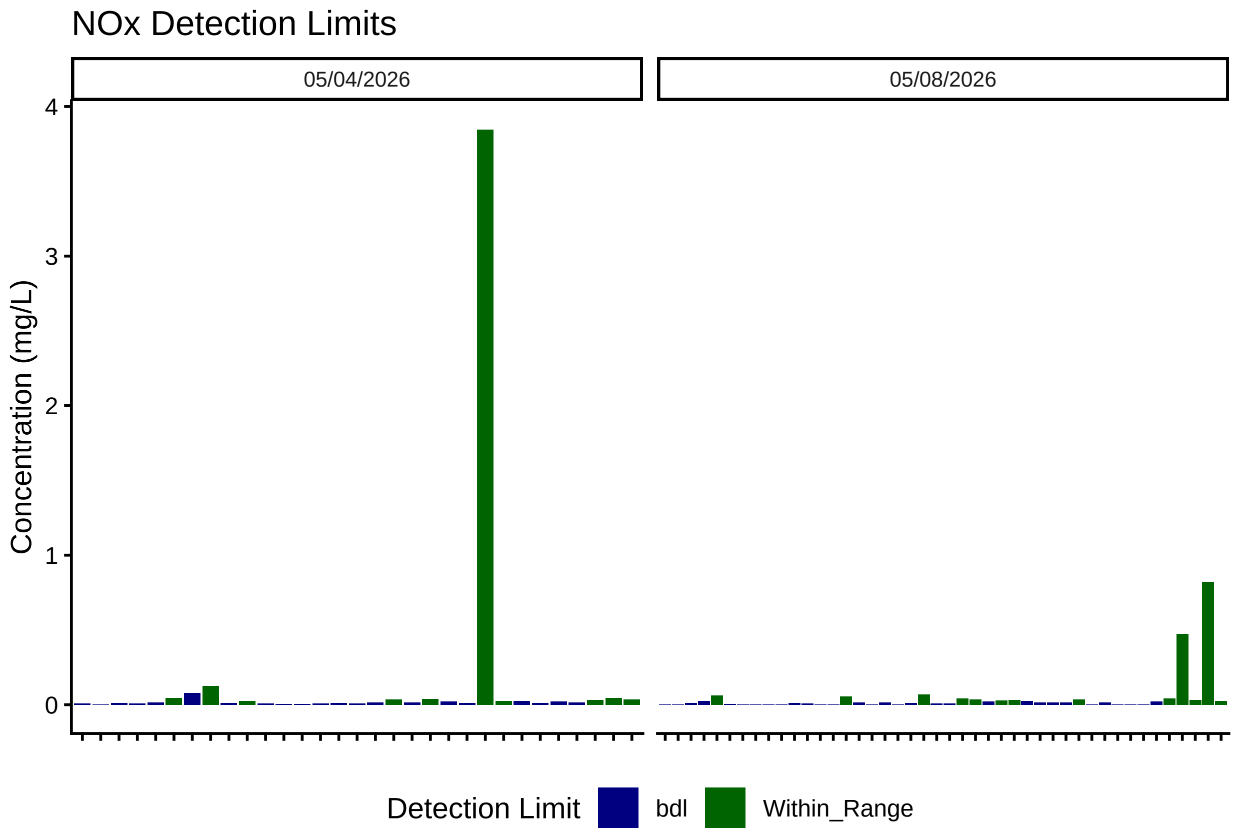


Test	Run_Date	Matrix_Flags
Ammonia 2	05/04/2026	Matrix Has Diff <50% - PROCEED
Ammonia 2	05/08/2026	Matrix Has Diff <50% - PROCEED
Vanadium NOx	05/04/2026	Matrix Has Diff <50% - PROCEED
Vanadium NOx	05/08/2026	Matrix Has Diff <50% - PROCEED
o-PHOS 0.3	05/04/2026	Matrix Has Diff <50% - PROCEED
o-PHOS 0.3	05/08/2026	Matrix Has Diff <50% - PROCEED

##Sample Flagging - Within range of standard curve




PO4 Detection Limits
05/04/2026
05/08/2026
Concentration (mg/L)
0.20
0.15
0.10
0.05
0.00
Detection Limit
bdl
Within_Range



##Add QAQC Flags

0.12 Check to see if samples run match metadata & merge info

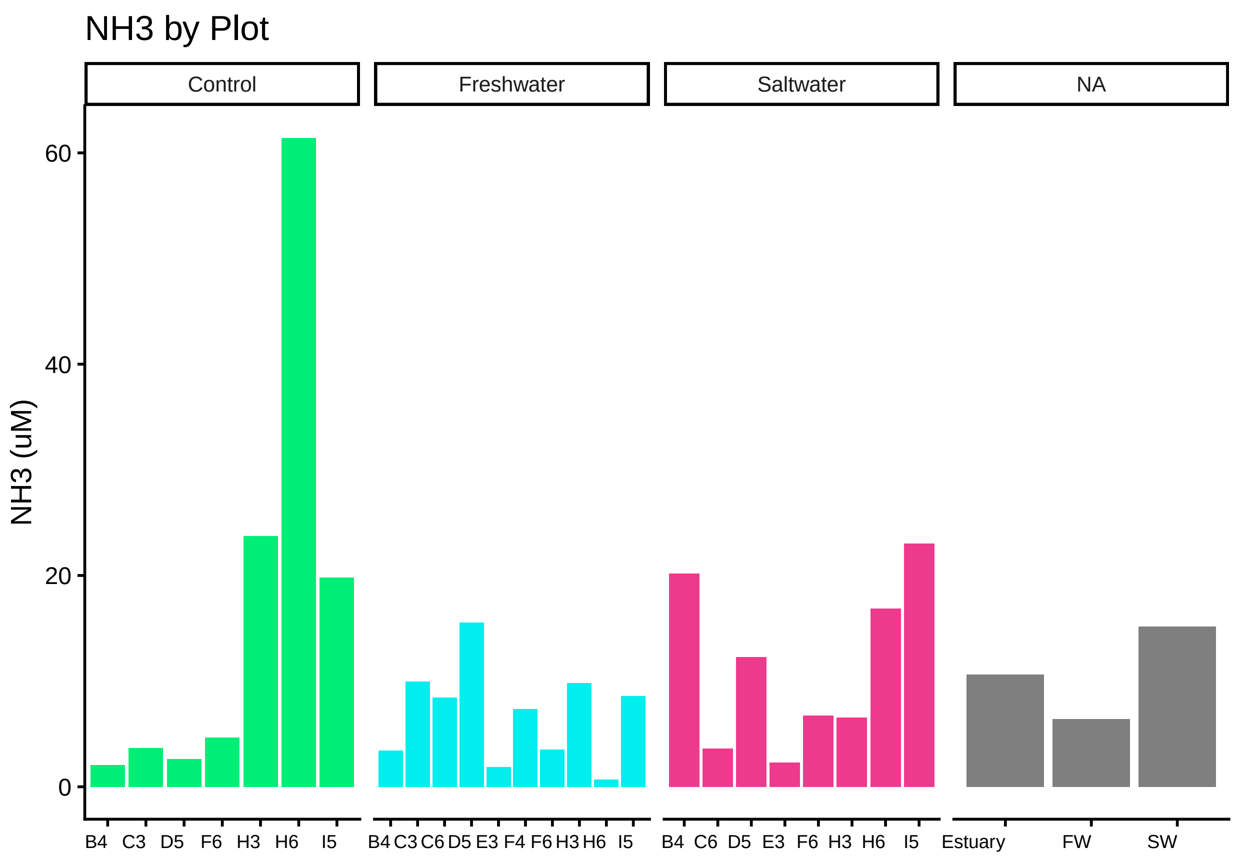
```
## Some sample IDs are missing from metadata.

## [1] "TMP_C_H6_20260602"

##  [1] "Sample_Name" "Run_Number"   "Conc"         "Absorbance"   "Dilution"
##  [6] "Unit"        "Test"         "Run_Time"     "Run_Date"     "Keep"
## [11] "Pair_ID"     "Conc_uM"      "Conc_flag"
```

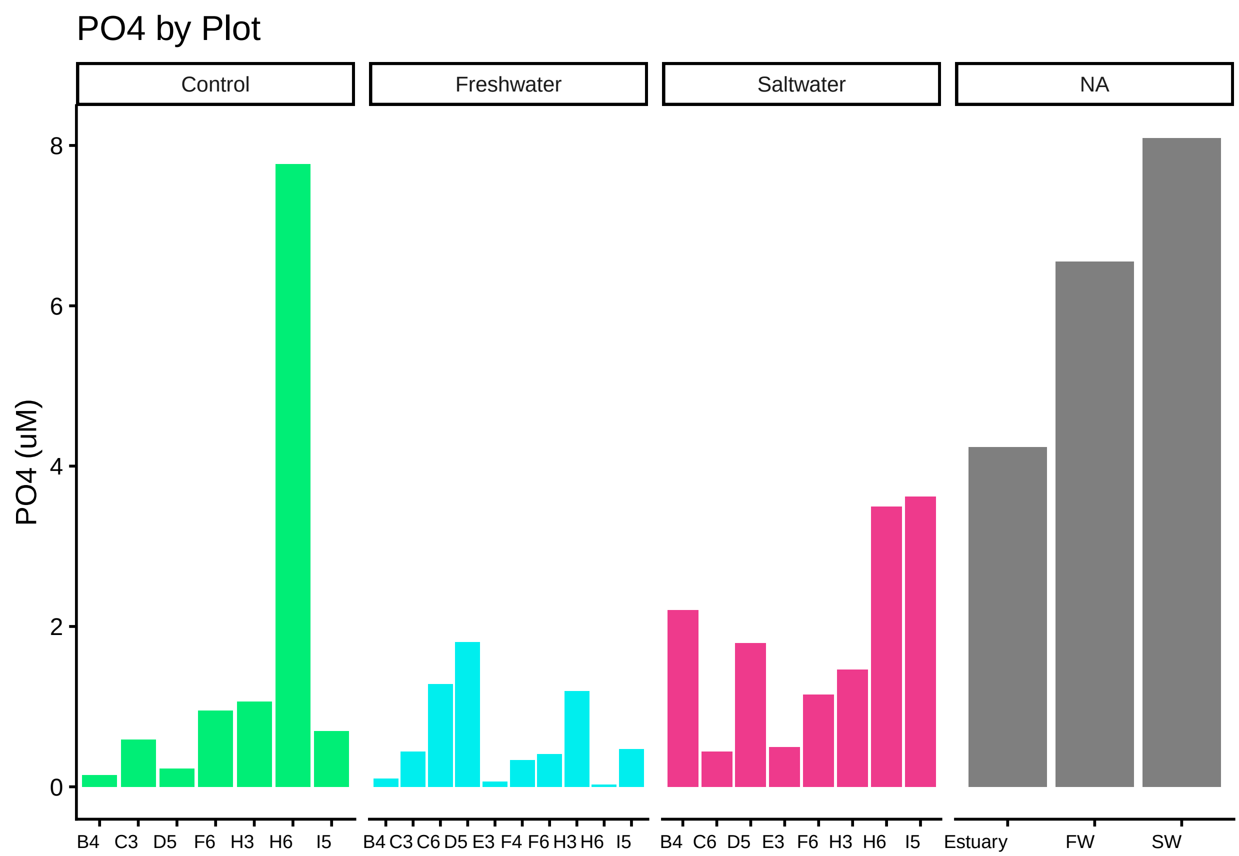
##Format Data

0.13 Visualize Data by Plot

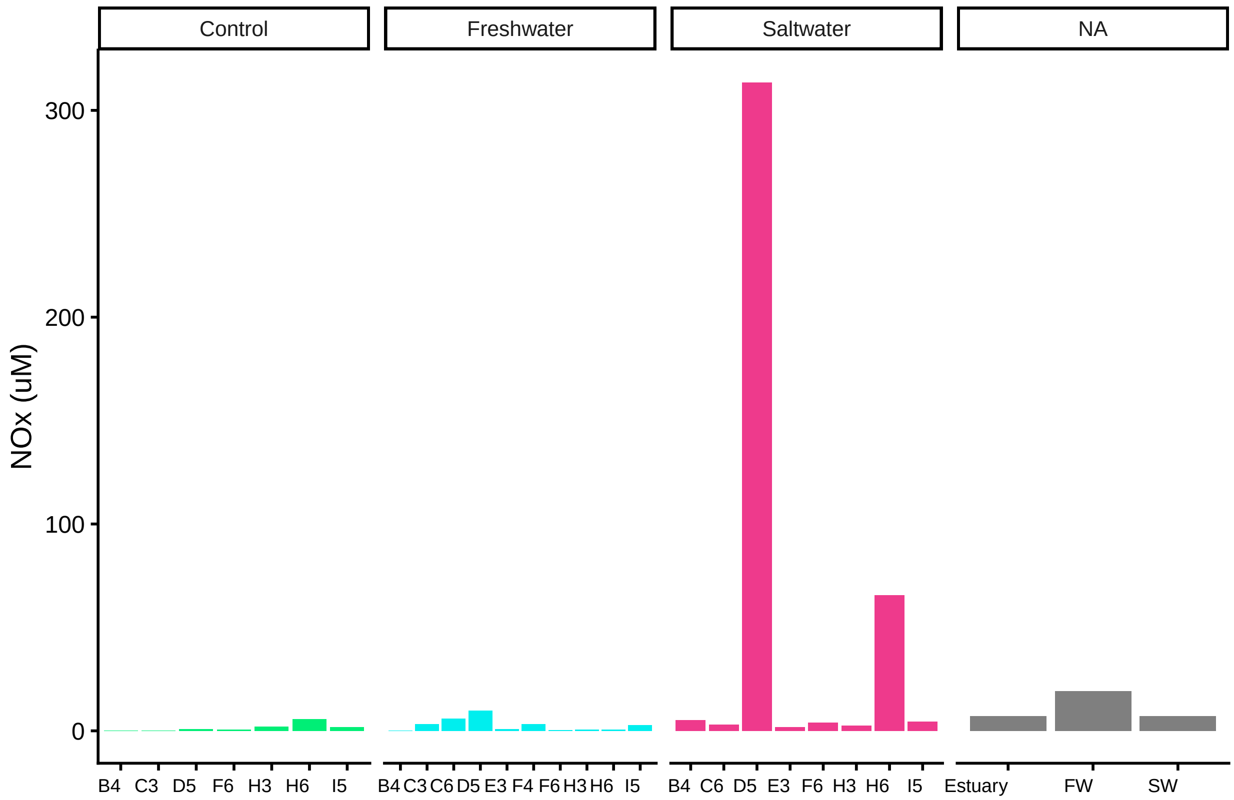
```
##   Site_Code Plot Grid_Square     Date
## 1       TMP    C          B4 20260309
## 2       TMP    C          C3 20260309
## 3       TMP    C          D5 20260309
## 4       TMP    C          F6 20250711
## 5       TMP    C          H3 20250711
## 6       TMP    C          H3 20260309
```

```
##   Site_Code Plot Grid_Square     Date         Sample_ID    Site Zone Grid
## 1       TMP    C          B4 20260309 TMP_C_B4_20260309 Tempest    C   B4
## 2       TMP    C          C3 20260309 TMP_C_C3_20260309 Tempest    C   C3
## 3       TMP    C          D5 20260309 TMP_C_D5_20260309 Tempest    C   D5
## 4       TMP    C          F6 20250711 TMP_C_F6_20250711 Tempest    C   F6
## 5       TMP    C          H3 20250711 TMP_C_H3_20250711 Tempest    C   H3
## 6       TMP    C          H3 20260309 TMP_C_H3_20260309 Tempest    C   H3
##   Depth_cm Event_Time Time_of_day Source Volume_mL Collection_Date
## 1       15       <NA>        <NA>     PW       9.0        20260309
## 2       15       <NA>        <NA>     PW       4.0        20260309
## 3       15       <NA>        <NA>     PW       5.0        20260309
## 4       15       <NA>        <NA>     PW       7.5        20250711
## 5       15       <NA>        <NA>     PW      14.0        20250711
## 6       15       <NA>        <NA>     PW       8.0        20260309
##   Evacuation_Date NOx_Conc_mgL NOx_Conc_uM NOx_Conc_Flag NOx_QAQC_Flag
## 1        20260306     0.002901   0.2071152           bdl
## 2        20260306     0.001754   0.1252258           bdl
## 3        20260306     0.012405   0.8856476           bdl
## 4        20250708     0.007791   0.5562338           bdl
## 5        20250708     0.002212   0.1579244           bdl
## 6        20260306     0.024258   1.7318855           bdl
##   NH3_Conc_mgL NH3_Conc_uM NH3_Conc_Flag NH3_QAQC_Flag PO4_Conc_mgL PO4_Conc_uM
## 1     0.028899    2.063227  Within_Range                   0.004516   0.1458008
## 2     0.051426    3.671529  Within_Range                   0.018202   0.5876587
## 3     0.037136    2.651303  Within_Range                   0.007024   0.2267726
## 4     0.065259    4.659127  Within_Range                   0.029424   0.9499653
## 5     0.077177    5.510006  Within_Range                   0.013632   0.4401144
## 6     0.255198   18.219709  Within_Range                   0.019172   0.6189755
##   PO4_Conc_Flag PO4_QAQC_Flag Analysis_rundate Run_notes Field_notes
## 1  Within_Range                     05/08/2026                  <NA>
## 2  Within_Range                     05/08/2026                  <NA>
## 3  Within_Range                     05/08/2026                  <NA>
## 4  Within_Range                     05/04/2026                  <NA>
## 5  Within_Range                     05/04/2026                  <NA>
## 6  Within_Range                     05/08/2026                  <NA>
```


NH3 by Plot
Control
Freshwater
Saltwater
NA
NH3 (uM)
60
40
20
0
B4
C3
D5
F6
H3
H6
I5
B4
C3
C6
D5
E3
F4
F6
H3
H6
I5
B4
C6
D5
E3
F6
H3
H6
I5
Estuary
FW
SW


PO4 by Plot
Control
Freshwater
Saltwater
NA
PO4 (uM)
0
2
4
6
8
B4 C3 D5 F6 H3 H6 I5
B4 C3 C6 D5 E3 F4 F6 H3 H6 I5
B4 C6 D5 E3 F6 H3 H6 I5
Estuary FW SW

NOx by Plot



##Write Out Data

#end